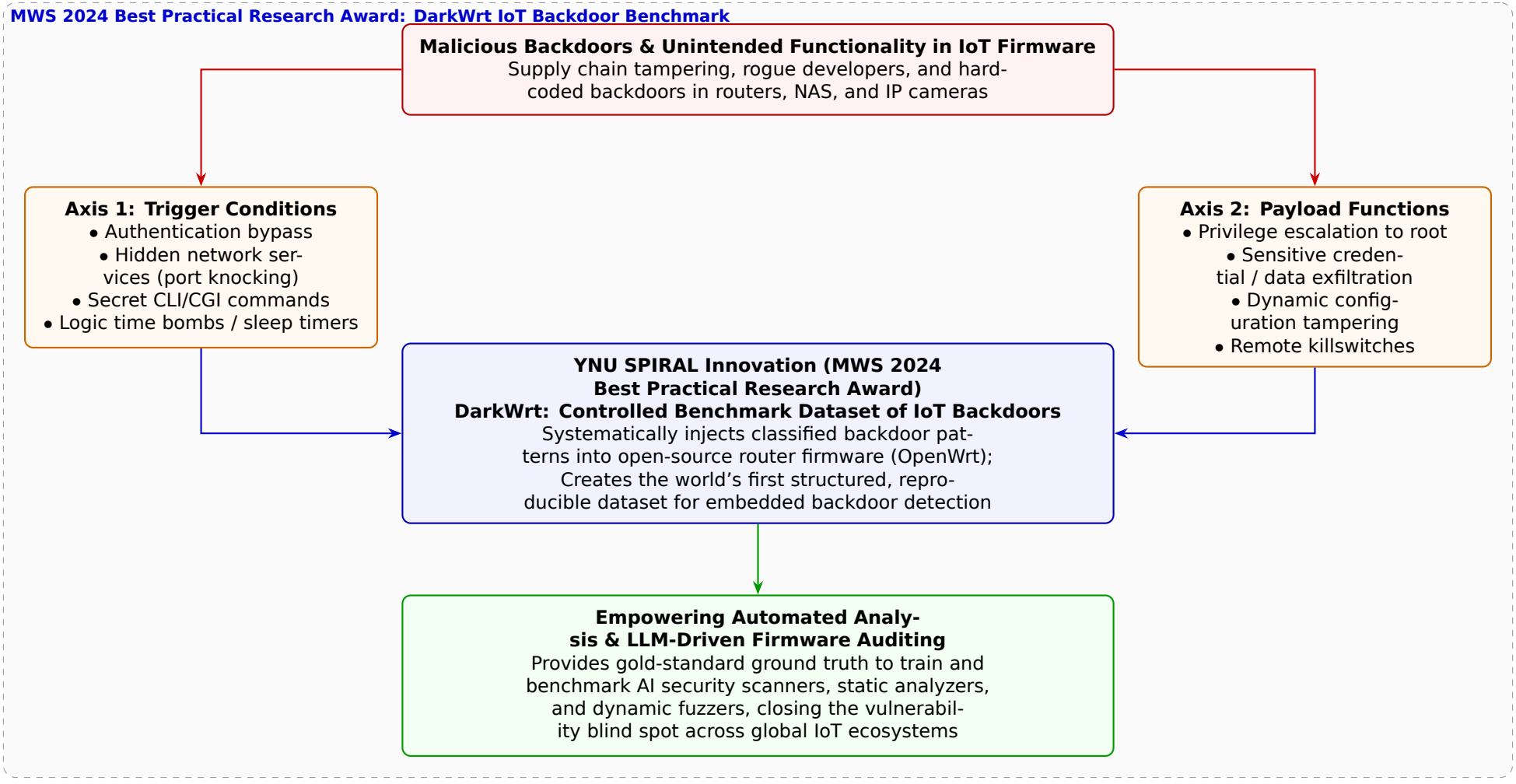
MWS 2024 Best Practical Research Award: DarkWrt IoT Backdoor Benchmark
Malicious Backdoors & Unintended Functionality in IoT Firmware
Supply chain tampering, rogue developers, and hard-coded backdoors in routers, NAS, and IP cameras
Axis 1: Trigger Conditions
• Authentication bypass
• Hidden network services (port knocking)
• Secret CLI/CGI commands
• Logic time bombs / sleep timers
Axis 2: Payload Functions
• Privilege escalation to root
• Sensitive credential / data exfiltration
• Dynamic configuration tampering
• Remote killswitches
YNU SPIRAL Innovation (MWS 2024 Best Practical Research Award)
DarkWrt: Controlled Benchmark Dataset of IoT Backdoors
Systematically injects classified backdoor patterns into open-source router firmware (OpenWrt); Creates the world's first structured, reproducible dataset for embedded backdoor detection
Empowering Automated Analysis & LLM-Driven Firmware Auditing
Provides gold-standard ground truth to train and benchmark AI security scanners, static analyzers, and dynamic fuzzers, closing the vulnerability blind spot across global IoT ecosystems